Solving the Model

Two countries $i \in \{US, R\}$, three sectors j, years $t = 1, \ldots, T$. A hat is counterfactual over baseline, $\widehat{x} = x'/x$. A prime is a counterfactual level. Anything with neither is a number you already have.

Twenty-three steps, in order. Each one is a commented block in the code, with the same number and the same name.

Part 1 — Build the baseline `baseline.m`

1. Value added.

$$VA_{i,t} = \sum_j \beta_i^j Y_{i,t}^j \tag{1}$$

2. Consumption expenditure and the investment rate.

$$E_{i,t}^C = VA_{i,t} - E_{i,t}^X - NX_{i,t}, \qquad x_{i,t} = E_{i,t}^X / VA_{i,t} \tag{2}$$

3. Consumption shares.

$$s_{i,t}^j = \frac{E_{i,t}^j - \sum_m \nu_i^{m,j}\left(1 - \beta_i^m\right) Y_{i,t}^m - \upsilon_i^j E_{i,t}^X}{E_{i,t}^C} \tag{3}$$

4. Phitilde, how much consumption a dollar buys.

$$\widetilde{\phi}_{i,t} = \Big(\sum_j \varepsilon^j s_{i,t}^j \Big)^{-1} \tag{4}$$

5. World interest rate, for $t = 1, \ldots, T-1$.

$$R_{t+1} = \frac{1}{\delta} \cdot \frac{E_{R,t+1}^C}{E_{R,t}^C} \cdot \frac{L_{R,t}}{L_{R,t+1}} \cdot \frac{\widetilde{\phi}_{R,t}}{\widetilde{\phi}_{R,t+1}} \tag{5}$$

6. The wedge, for $t = 1, \ldots, T-1$. ROW's is one.

$$\omega_{US,t} = \frac{L_{US,t+1}}{L_{US,t}} \cdot \frac{\widetilde{\phi}_{US,t+1}}{\widetilde{\phi}_{US,t}} \cdot \delta R_{t+1} \cdot \frac{E_{US,t}^C}{E_{US,t+1}^C} \tag{6}$$

7. Bonds. Start from $B_{i,1} = 0$.

$$B_{i,t+1} = R_t B_{i,t} + NX_{i,t} \tag{7}$$

Part 2 — One year `solveyear.m`, `yearmodel.m`

You are handed the year t and US net exports $NX'_{US,t}$. Set $NX'_{R,t} = -NX'_{US,t}$.

8. Guess the change in US income, $\widehat{VA}_{US}$. Set $\widehat{VA}_R = 1$.

9. Unit costs and prices. Start at $\widehat{P} = 1$ and alternate the two until they stop moving.

$$\widehat{c}_i^j = \left(\widehat{VA}_i\right)^{\beta_i^j} \prod_m \left(\widehat{P}_i^m\right)^{\nu_i^{j,m}(1-\beta_i^j)}, \qquad \widehat{P}_i^j = \Big[\sum_h \pi_{ih,t}^j \left(\widehat{c}_h^j\right)^{-\theta}\Big]^{-1/\theta} \tag{8}$$

10. Trade shares.

$$\pi'^j_{ih} = \pi^j_{ih,t}\left(\frac{\widehat{c}_h^j}{\widehat{P}_i^j}\right)^{-\theta} \tag{9}$$

11. Income and spending.

$$VA'_i = VA_{i,t}\widehat{VA}_i, \qquad E_i^{X\prime} = x_{i,t}VA'_i, \qquad E_i^{C\prime} = \big(1 - x_{i,t}\big)VA'_i - NX'_i \tag{10}$$

12. Real consumption. Solve for $\widehat{C}_i$, one country at a time, where $\widehat{E}_i^C = E_i^{C\prime}/E_{i,t}^C$.

$$\widehat{C}_i \left[\sum_j s_{i,t}^j \left(\widehat{P}_i^j\right)^{1-\psi} \widehat{C}_i^{(1-\psi)(\varepsilon^j-1)}\right]^{\frac{1}{1-\psi}} = \widehat{E}_i^C \tag{11}$$

13. New consumption shares. The bracket above, to the same power, is $\widehat{P}_i^C$.

$$s'^j_i = s^j_{i,t}\left(\frac{\widehat{P}_i^j}{\widehat{P}_i^C}\right)^{1-\psi} \widehat{C}_i^{(1-\psi)(\varepsilon^j-1)} \tag{12}$$

14. Market clearing. Six slots, ordered $(a,US),(a,R),(m,US),(m,R),(s,US),(s,R)$.

$$F_{(j,i)} = s'^j_i E_i^{C\prime} + v_i^j E_i^{X\prime}, \quad \Gamma_{(j,i),(m,i)} = \nu_i^{m,j}\big(1-\beta_i^m\big), \quad \Pi'_{(j,i),(j,h)} = \pi'^j_{hi} \tag{13}$$

$$E' = \big[I - \Gamma\Pi'\big]^{-1} F, \qquad Y' = \Pi' E' \tag{14}$$

15. Income residual. If this is not zero, go back to step 8.

$$g = \frac{\sum_j \beta_i^j Y'^j_{US}}{VA_{US,t}\,\widehat{VA}_{US}} - 1 \tag{15}$$

Part 3 — All years `solvepath.m`, `pathmodel.m`

You are handed the wedge path $\omega_{US,t}$.

16. Guess the whole path $\{NX'_{US,t}\}_{t=1}^{T}$. Start at the observed one.

17. Every year. Run Part 2 at each t, collect $E_{i,t}^{C\prime}$ and the new shares, then recompute step 4 from those shares.

18. World interest rate, step 5 again, on the new consumption path.

19. Bonds, step 7 again, on the guessed net exports and the new interest rates.

20. US saving equation. For $t = 1, \ldots, T-1$,

$$\mathcal{R}_t = \ln \frac{E_{US,t+1}^{C\prime}}{E_{US,t}^{C\prime}} - \ln \frac{L_{US,t+1}}{L_{US,t}} - \ln \frac{\widetilde{\phi}_{US,t+1}}{\widetilde{\phi}_{US,t}} - \ln \frac{\delta R'_{t+1}}{\omega_{US,t}}, \tag{16}$$

21. Terminal condition, so the counterfactual ends where the baseline did.

$$\mathcal{R}_T = \frac{B'_{US,T+1} - B_{US,T+1}}{VA_{US,T}}. \tag{17}$$

22. Newton. If the residuals are not zero, adjust the path and go back to step 16.

Part 4 — The experiment `script_experiment.m`

23. The experiment. Run Part 3 twice — once at the wedge from step 6, once with the wedge set to one — and compare manufacturing's share of value added,

$$va_{i,t}^{j} = \frac{\beta_i^j Y_{i,t}^{\prime j}}{\sum_m \beta_i^m Y_{i,t}^{\prime m}}, \qquad \text{answer} = 1 - \frac{\Delta va_{US}^{m,\text{CF}}}{\Delta va_{US}^{m,\text{base}}}, \tag{18}$$

with Δ the change over the sample. The first run must reproduce the data. Compare the two runs against each other, never against the data.

What the numbers are, and where they come from

	Meaning	Column
β_i^j	value added share of output	`recovered_constants.beta`
υ_i^j	investment shares	`recovered_constants.upsilon`
$\nu_i^{j,m}$	sector j's spending share on good m	`recovered_nu.nu`
$\pi_{ih,t}^j$	buyer i's share sourced from seller h	`ImpExpSectorPanel.imp_sh`
$Y_{i,t}^j$	gross output	`CountrySectorPanel.go_wiot`
$E_{i,t}^j$	sectoral expenditure	`CountrySectorPanel.expend`
$E_{i,t}^X$	investment	`CountryPanel.tot_invest`
$NX_{i,t}$	net exports	`CountryPanel.tot_nx`
$L_{i,t}$	employment	`CountryPanel.emp`
$\theta = 4$	trade elasticity	
$\psi = 0.57$	consumption elasticity	
$\delta = 0.95$	discount factor	
ε^j	income elasticities	(0.0091, 0.4566, 0.5342)